Shapley value(s)

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Game Theory and Mechanism Design - Lecture 13

(This lecture heavily relies on Narahari (2014) Game Theory and Mechanism Design – chap. 29)

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- Understand how different games lead to different **coalition formation**
- Define and compute **Shapley value(s)**
- Discuss **properties** of Shapley value(s)

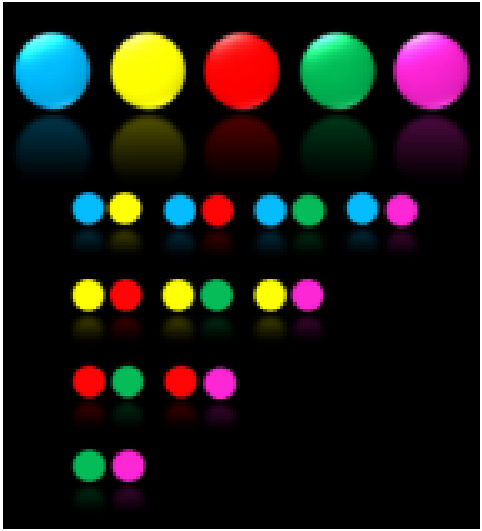
Outline

1. Which coalition will form?
2. How should they share?
3. Shapley value(s)
4. Shapley value(s) and the core

1. **Which coalition will form?**

Which one of the 2^N?

We have defined the power set 2^N as the set of all potential coalitions...



For non-cooperative games, we have worked towards finding an *outcome* (in terms of pay-offs), relying on some form of equilibrium...

How to define an outcome for a coalitional game?
(also referred to as cooperative game)

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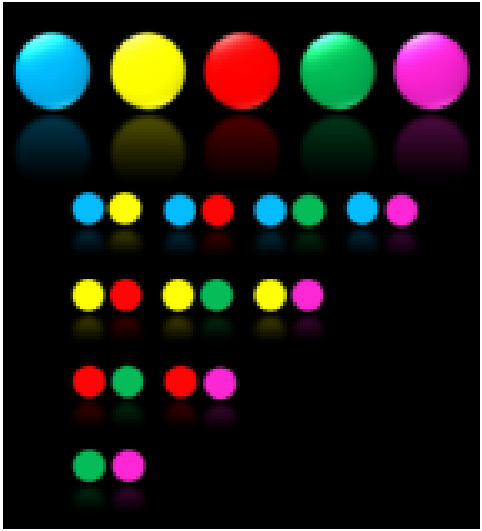

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How to define an outcome for a coalitional game?
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- we should identify which coalitions could be those that prevail
- we want to define and understand the distribution of pay-offs among players (the so-called *allocations*)

Monotonic games

Let us first look at the case of a *monotonic* game (but not super-additive, and hence not convex)...

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Coursework with rewards game (monotonic-only version)

Consider the following characteristic function:

$v(\emptyset) = 0$	$v(\{1\}) = 50$	$v(\{1,2\}) = 80$	$v(\{1,2,3\}) = 150$
	$v(\{2\}) = 55$	$v(\{2,3\}) = 85$	
	$v(\{3\}) = 65$	$v(\{1,3\}) = 95$	

Which coalitions do you believe can come out of that game?

- Is there any allocation that could make all players agree on $\{1,2,3\}$?
- Is there any allocation that could make players 1 and 2 agree on $\{1,2\}$? (same for the other 2-player coalitions)
- Is there any allocation that could make all players agree on being on their own?

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So, the outcome will be for them to be on their own!

Note that this result is for this game only... other non-super-additive games may still be advantageous for some coalitions.

Super-additive games

We then look at the case of a *super-additive* game (but not convex)...

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- Would specific allocations dominate others?

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Convex games

We finally look at the case of a *convex* game...

Coursework with rewards game (convex version)

Consider the following characteristic function:

$v(\emptyset) = 0$	$v(\{1\}) = 30$	$v(\{1,2\}) = 85$	$v(\{1,2,3\}) = 200$
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	$v(\{3\}) = 40$	$v(\{1,3\}) = 90$	

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- Would specific allocations dominate others?

So, the outcome will be for them to want to be all together! (this was obvious from the start, since convex games are also super-additive)

Super-additivity and the grand coalition

We like games where there is a strong incentive to be in it all together!

Definition (grand coalition)

Given a coalitional game (N, v) and potential coalitions $\mathcal{C} \subseteq N$, we call the *grand coalition* the specific coalition with all players together, i.e., $\mathcal{C} = N$.

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The reason why the grand coalition is important is that for many games (of interest), it is the one coalition that matters...

Proposition (super-additive games and the grand coalition)

If a coalitional game (N, v) is super-additive, the only coalition $\mathcal{C} \subseteq N$ that makes sense as an outcome is the grand coalition $\mathcal{C} = N$. Since convex games are also super-additive, the result holds for them too.

A practical benefit is that we do not have to enumerate over all potential coalitions.

2. **How should they share?**

Please, tell me what you think...

Some basic notations

In previous lectures, we mentioned that we have allocations $x = (x_1, \ldots, x_n)$.
These are the result of an *allocation process*.

Definition (allocation process)

For a coalitional game (N, v), an *allocation process* is a function

$$\psi : \mathbb{N} \times \mathbb{R}^{|2^N|} \to \mathbb{R}^{|N|}$$

where

- $\psi(N, v)$ is the vector of pay-offs for all players (an allocation)
- $\psi_i(N, v)$ is the pay-off of player i

To lighten notations, we usually write $\psi(v)$ and $\psi_i(v)$ only.

Now, how should we define ψ for the coalition's sharing of pay-offs to be

- **stable** (stability comes from the core – see previous lecture), and
- **fair**

We will define fairness through a number of *axioms*.

Efficient

Whatever allocation is considered, it should be Pareto efficient (link to collective rationality).

Axiom (efficiency)

Given a coalitional game (N, v), the efficiency axiom states that, to be fair to all players, all of the joint pay-offs should be allocated, i.e.,

$$\sum_{i \in N} \psi_i(v) = v(N)$$

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For this game, we will necessarily have $\psi_1(v) + \psi_2(v) + \psi_3(v) = 210$.

Dummy player

The *dummy player* is the one that does not bring anything into the game.

Definition (dummy player)

Given a coalitional game (N, v), a *dummy player* i is one for which pay-offs do not increase when they join any coalition $\mathcal{C} \subseteq N$, i.e.

$$v(\mathcal{C} \cup \{i\}) = v(\mathcal{C}), \ \forall \mathcal{C} \subseteq N$$

Axiom (dummy player)

Given a coalitional game (N, v), the dummy player axiom states that it is only fair the dummy player i gets 0:

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Coursework with rewards game (with dummy player)

Consider the following characteristic function:

$v(\emptyset) = 0$	$v(\{1\}) = 0$	$v(\{1,2\}) = 55$	$v(\{1,2,3\}) = 145$
	$v(\{2\}) = 55$	$v(\{2,3\}) = 145$	
	$v(\{3\}) = 60$	$v(\{1,3\}) = 60$	

Can you identify the dummy player?

Symmetry

If they consistently contribute the same, their share should be the same.

Definition (interchangeable player)

Given a coalitional game (N, v), players i and j are said to be interchangeable is

$$v(\mathcal{C} \cup \{i\}) = v(\mathcal{C} \cup \{j\}), \ \forall \mathcal{C} \subseteq N, \quad i, j \notin \mathcal{C}$$

Axiom (symmetry axiom)

Given a coalitional game (N, v), the symmetry axiom states that it is only fair for interchangeable players i and j to receive the same share in the allocation, i.e.

$$v(\mathcal{C} \cup \{i\}) = v(\mathcal{C} \cup \{j\}), \ \forall \mathcal{C} \subseteq N, i, j \notin \mathcal{C} \quad \Longrightarrow \quad \psi_i(v) = \psi_j(v)$$

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Coursework with rewards game (with interchangeable players)

Consider the following characteristic function:

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	$v(\{2\}) = 65$	$v(\{2,3\}) = 140$	
	$v(\{3\}) = 50$	$v(\{1,3\}) = 120$	

We then expect that $\psi_1(v) = \psi_3(v)$

Linearity (and additivity)

Axiom (symmetry axiom)

Given two coalitional games (N, v_1) and (N, v_2), and a weight $\alpha \in [0, 1]$, define a new game (N, w) such that

$$w = \alpha v_1 + (1 - \alpha) v_2$$

Then, the linearity axiom states that it is only fair that the pay-off allocation follows the same linear combination relation, i.e.

$$\psi_i(w) = \alpha \psi_i(v_1) + (1 - \alpha) \psi_i(v_2), \quad \forall i$$

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Multiple coursework with rewards game

Consider the 2 following characteristic functions (without empty sets, as these are 0s anyway):

$v_1(\{1\}) = 50$, $v_2(\{1\}) = 30$ $\quad$ $v_1(\{1,2\}) = 120$, $v_2(\{1,2\}) = 100$

$v_1(\{2\}) = 55$, $v_2(\{2\}) = 25$ $\quad$ $v_1(\{2,3\}) = 136$, $v_2(\{2,3\}) = 116$ $\quad$ $v_1(\{1,2,3\}) = 210$, $v_2(\{1,2,3\}) = 190$

$v_1(\{3\}) = 65$, $v_2(\{3\}) = 40$ $\quad$ $v_1(\{1,3\}) = 144$, $v_2(\{1,3\}) = 144$

I decide to define a new game saying that $w = \frac{1}{2}(v_1 + v_2)$...

... then, in terms of fair allocation, I should have $\psi_i(w) = \frac{1}{2}(\psi_i(v_1) + \psi_i(v_2))$ $(i = 1, 2, 3)$.

3. **Shapley value(s)**

Shapley's idea

The pay-off allocation should be based on the marginal contribution of each agent to the coalition

Definition (marginal contribution)

Given a coalitional game (N, v), the *marginal contribution* $\delta(i, \mathcal{C})$ of player i to a coalition $\mathcal{C} \subseteq N$ they are part of, is the difference in overall pay-off *with* and *without* that player, i.e.,

$$\delta(i, \mathcal{C}) = v(\mathcal{C}) - v(\mathcal{C} \setminus i)$$

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We then have

$\delta(1, \{1\}) = 50$	$\delta(1, \{1,2\}) = 65$	$\delta(2, \{1,2\}) = 70$	$\delta(1, \{1,2,3\}) = 74$
$\delta(2, \{2\}) = 55$	$\delta(2, \{2,3\}) = 71$	$\delta(3, \{2,3\}) = 81$	$\delta(2, \{1,2,3\}) = 66$
$\delta(3, \{3\}) = 65$	$\delta(1, \{1,3\}) = 79$	$\delta(3, \{1,3\}) = 94$	$\delta(3, \{1,2,3\}) = 90$

So, eventually, which marginal contribution should we consider?

The Shapley value

Since marginal contribution changes as a function of the coalition, we should consider the average marginal contribution over all coalitions and permutations.

Definition (Shapley value)

The Shapley value ψ_i is the pay-off to be given to player $i \in N$ for their average marginal contribution over all possible coalitions, i.e.

$$\psi_i = \frac{1}{n} \sum_{\mathcal{C} \subseteq N \setminus i} \binom{n-1}{|\mathcal{C}|}^{-1} \delta(i, \mathcal{C} \cup i)$$

where $|\mathcal{C}|$ is the cardinality of $\mathcal{C}$ and $\binom{n-1}{|\mathcal{C}|}$ denotes the number of coalitions of size $|\mathcal{C}|$ one could obtain with $n-1$ players.

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Coursework with rewards game (focusing on player 1 only)

There are 4 potential coalitions $\mathcal{C}$ that do not include player 1, $\{\emptyset\}, \{2\}, \{3\}, \{2,3\}$, yielding

$$\begin{aligned}
&\delta(1, \emptyset \cup \{1\}) = 50, && \text{with weight } 1/3 \\
&\delta(1, \{2\} \cup \{1\}) = 65, && \text{with weight } 1/6 \\
&\delta(1, \{3\} \cup \{1\}) = 79, && \text{with weight } 1/6 \\
&\delta(1, \{2,3\} \cup \{1\}) = 74, && \text{with weight } 1/3
\end{aligned} \quad \longrightarrow \quad \psi_1(v) = 65.33$$

And using the same procedure, for the other 2 players, $\psi_2(v) = 63.83$ and $\psi_3(v) = 80.83$.

4. **Shapley value(s) and the core**

I said **fair**... and **stable!**

In the core?

Shapley values are fair, but they may not be stable...

Proposition (Shapley values and the core)

In the general case, for a coalitional game (N, v), there is no guarantee that the allocation $x = (x_1, \ldots, x_n)$ defined by Shapley values $\psi_i(v)$ $(i = 1, \ldots, n)$ lies in the core $\mathbb{C}(N, v)$ of the game.

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Proposition (convex games, Shapley values, and the core)

For a coalitional game (N, v), if the game is convex, then the allocation $x = (x_1, \ldots, x_n)$ defined by Shapley values $\psi_i(v)$ $(i = 1, \ldots, n)$ lies in the core $\mathbb{C}(N, v)$ of the game.

And this is why the family of convex games is particularly interesting.

Wrapping up

- Shapley values comprise one of the **most celebrated concepts** in cooperative game theory
- **Fairness** and **stability** motivate its proposal (though stability is not always guaranteed, eventually!)
- Fairness is defined based on a set of **mathematical axioms** (are they enough, though?)
- Shapley values may not always be in the core...
- There exists **other solution concepts** for cooperative games (e.g., Nucleolus) which we will not cover.

Further readings (/lecture notes for this topic):

Narahari Y (2014) *Game Theory and Mechanism Design* - chap. 29 (available on Blackboard Learn)

Thanks for your attention!